MILITARY SPECIFICATION

POWER SUPPLY: 10559345 (PFN CHARGE)

1. SCOPE

1.1 This specification covers one type of electronic assembly known as Power Supply: 10559345, (PFN Charge) (see 6.1).

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

SPECIFICATIONS

Military

MIL-F-13926	Fire Control Materiel; General Specification Governing the Manufacture and Inspection of
MIL-P-14232	Parts, Equipment and Tools for Army Materiel: Packaging and Packing of
MIL-I-45607	Inspection Equipment, Supply and Maintenance of

STANDARDS

Military

MIL-STD-105	Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-109	Quality Assurance Terms and Definitions
MIL-STD-810	Environmental Test Methods

DRAWINGS

U.S. Army, Frankford Arsenal

F10559345	PFN Charge - Power Supply

FSC 1240

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PACKAGING DATA SHEET

10559345 — Packaging of Power Supply - (PFN Charge): 10559345

(Copies of specifications, standards, drawings, and packaging data sheets required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 Fabrication.- The power supply shall be manufactured in accordance with Drawing F10559345 and drawings pertaining thereto.

3.2 General specification.- The power supply shall meet the following requirements, where applicable, of MIL-F-13926:

(a) Order of precedence
(b) Dimensions and tolerances
(c) Part identification and marking

3.2.1 Electronic workmanship.- The power supply shall meet the workmanship requirements of Drawing F10559345 and drawings pertaining thereto.

3.3 Performance.- Unless otherwise specified, the power supply shall meet the performance requirements of this specification at standard ambient temperatures between 60 degrees Fahrenheit (°F) and 90°F.

3.3.1 Loads, power, and signals.- The power supply shall perform as specified herein when the loads, power, and signals of Table I are applied as specified.

TABLE I

ITEM	CONDITION	CHARACTERISTICS	CONNECTIONS
1.	Loads:		Connected between the following terminals:
1.1	Capacitor	160 μf, ±10%, 2 kV	P1-A-1 and P1-9
1.2	Capacitor	.22 μf, ±10%, 600V	P1-10 and P1-9
1.3	Resistor	3K, ±5%	P1-11 and P1-16
1.4	Resistor	3K, ±5%	P1-14 and P1-16
1.5	Resistor	66.5K, ±1%	As specified herein

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TABLE I (continued)

ITEM	CONDITION	CHARACTERISTICS	CONNECTIONS
1.6	Resistor	100K, ±1%	P1-6 and P1-9
1.7	Resistor	3.32 MΩ, ±1%	As specified herein
1.8	Resistor	32.4K, ±1%	As specified herein
2.	Power sources:	Tol. (V) / Max. ripple (mv, p-p)	Connected between the following terminals:
2.1	+5V	±.2 / 25	P1-16(+) and P1-9(-)
2.2	+15V	±.7 / 25	P1-15(+) and P1-9(-)
2.3	-6V	±.2 / 25	P1-12(-) and P1-9(+)
3.	Signal sources:		Applied to input terminals as specified herein
3.1	Digital type-A	Logical one: 4 ±1V Logical zero: .2 ±.2 V	
3.2	24 Vdc	+17 to +26V, adjustable	
3.3	8 Vdc	Tolerance: ±.5V Source impedance: 33K, ±5%	

3.3.1.1 Malfunction 6.- With the logical zero type-A signal of Table I applied to P1-4 (P1-9 return), the output at P1-8 shall be 4.2 ±1 Vdc. With the load (66.5 kilohms) of Table I connected between P1-5 and P1-15, the output at P1-8 shall be .2 ±.2 Vdc.

3.3.1.2 PFN voltage.- With the type-A digital signal of Table I applied to the input terminals as specified below, the performance shall be as specified in the following subparagraphs.

Input Terminals	Type-A Signal
P1-2	Logical zero
P1-3 and P1-4	Momentary logical zero

3.3.1.2.1 PFN voltage output.- With the 24 Vdc of Table I, adjusted to 24 ±1 Vdc, applied to P1-19 and P1-20 with the negative returns to P1-17 and P1-18, the PFN output voltage shall appear at P1-A-1.

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3.3.1.2.2 Dump relay.- With the PFN voltage at P1-A-1 set to 1200 ±100 Vdc by means of R10, removal of the 24 Vdc from P1-19 and P1-20 shall cause the voltage at P1-A-1 to drop to less than 50 volts within 25 seconds.

3.3.1.2.3 PFN charge time.- With R10 set as in 3.3.1.2.2, reapplying the 24 Vdc of Table I to P1-19 and P1-20 at a voltage level of 24 ±1 Vdc and 18.5 ±.5 Vdc shall cause the voltage at P1-A-1 to increase to 1200 ±100 Vdc in less than 5 seconds, and in less than 6 seconds at the temperature extremes of 3.3.2.2. The repeaking circuit shall maintain the PFN voltage at this steady state value within 15 volts p-p.

3.3.1.2.4 Repeaking inhibit.- With the 8-Vdc signal of Table I applied to P1-7, the repeaking circuit shall be inhibited, thereby disabling the PFN charge circuit. With the 8-Vdc signal removed from P1-7, and the type-A signal at P1-2 changed to logical one, the repeaking circuit shall be inhibited. With the type-A signal at P1-2 changed to logical zero and the load (66.5 kilohms) of Table I connected between P1-5 and P1-15, the repeaking circuit shall be inhibited.

3.3.1.2.5 PFN voltage range adjustment.- The PFN voltage at P1-A-1 shall be capable of being set to 1000 ±20 Vdc and 1500 ±30 Vdc by adjustment of R10.

3.3.1.2.6 PFN voltage programming.- With the PFN voltage (P1-A-1) set to 1200 ±15 Vdc, changing the load between P1-6 and P1-9 from 100 kilohms to 32.4 kilohms shall change the PFN voltage to 1345 ±40 Vdc, and to 1345 ±95 Vdc at the temperature extremes of 3.3.2.2. Changing this load from 32.4 kilohms to 3.32 megohms shall change the PFN voltage to 1030 ±30 Vdc, and 1030 ±71 Vdc at the temperature extremes of 3.3.2.2.

3.3.1.2.7 15-Vdc interlock.- Removal of the 15 Vdc from P1-15 shall inhibit the PFN charge circuit.

3.3.1.2.8 400-volt output.- With 24 ±1 Vdc applied to P1-19 and P1-20, the output at P1-10 shall increase to 400 ±30 volts, and to 400 ±60 volts at the temperature extremes of 3.3.2.2, in less than 6 seconds.

3.3.1.2.9 PFN voltage sensing.- With the PFN voltage at P1-A-1 set to 1200 ±25 volts, the current through a microammeter load between P1-5 and P1-9 shall be 75 ±5 microamperes.

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3.3.1.3 Fire signal.

3.3.1.3.1 With the signals of Table I applied to the input terminals as specified below (24 Vdc applied last), and with the PFN voltage at P1-A-1 adjusted to 1200 ±200 volts, the output at P1-11 shall be 4.2 ±1 volts.

INPUT TERMINALS	APPLIED SIGNAL
P1-2	Logical zero, Type-A
P1-3 and P1-4	Momentary logical zero, Type-A
P1-19 and P1-20	24 ±1 Vdc

3.3.1.3.2 With the PFN output voltage present (24 ±1 Vdc applied) and adjusted to 1200 ±200 volts, and with the input conditions of Table II applied sequentially, the outputs at P1-11 shall be as specified in Table II.

TABLE II

ITEM	INPUT PINS OF P1						OUTPUT
	13	1	3	4	2	7	P1-11
1	1	1	1	1	0	A	.2 ±.2 volt
2	0	1	1	1	1	B	4.2 ±1 volts
3	1	1	1	1	1	B	.2 ±.2 volt
4	1	1	1	1	0	B	.2 ±.2 volt
5	1	0	1	1	1	B	4.2 ±1 volts
6	1	1	1	0	1	B	.2 ±.2 volt
7	1	1	0	1	0	A	4.2 ±1 volts

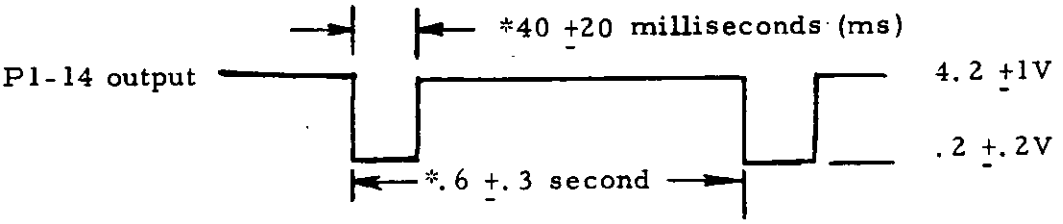
Legend: 1 = logical one, type-A A = 8 Vdc (per Table I)
0 = logical zero, type-A B = Open circuit

3.3.1.4 Range light.

3.3.1.4.1 With a momentary logical zero, type-A signal applied to P1-3, the output at P1-14 shall be 4.2 ±1 Vdc.

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3.3.1.4.2 With a logical zero, type-A signal applied to P1-2, a momentary logical zero, type-A signal applied to P1-4, and 24 ±1 volts applied to P1-19 and P1-20, the waveform of figure 1 shall appear at P1-14 after at least 6 seconds have elapsed.



*Under the temperature extremes of 3.3.2.2 the pulse width shall be 40 ±32 ms, and the pulse spacing .6±.5 second.

Figure 1 - P1-14 Output

3.3.1.4.3 With a momentary logical zero, type-A signal applied to P1-1, the output at P1-14 shall be 0.2 ±.2 Vdc.

3.3.1.5 Current limiting of power driver.- With a momentary logical zero type-A signal applied to P1-3 and P1-4, and a short circuit between P1-A-1 and P1-9, the average current of the 24-volt signal when applied to P1-19 and P1-20 shall be less than 4 amperes. (Caution: The capacitor load between P1-A-1 and P1-9 shall have been discharged before short circuit is applied. The short circuit condition shall be limited to 5 seconds.)

3.3.2 Environmental.

3.3.2.1 Storage temperature.- The power supply shall meet the requirements of 3.3.1 at ambient temperature (60°F to 90°F) after exposure and thermal stabilization at -80°F and +160°F.

3.3.2.2 Operating temperatures.- The power supply shall meet the requirements of 3.3.1 while exposed to and thermally stabilized at -40°F and +125°F, subsequent to which it shall meet the requirements of 3.3.1 at ambient temperature (60°F to 90°F).

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3.3.2.3 Vibration.- The power supply shall meet the requirements of 3.3.1 after being exposed to the following vibratory conditions:

(a)	Amplitude:	0.5 inch maximum
(b)	Acceleration:	4 g maximum
(c)	Crossover frequency:	12.5 Hertz
(d)	Sweep time:	5 to 500 to 5 Hertz in 15 minutes. Two sweeps in each of the three mutually perpendicular axes.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection.- Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 General provisions.- The component and subassembly inspection requirements of MIL-F-13926 form a part of the quality assurance provisions of this specification. Definitions of inspection terms shall be as listed in MIL-STD-109.

4.2 First article (initial production) approval.- The requirement for first article approval and the responsibility (Government or contractor) for first article testing shall be as specified in the contract. The sample for first article approval tests shall consist of three power supplies. The sample shall be manufactured in the same manner, using the same materials, equipment, processes, and procedures as used in regular production. All parts and materials, including packaging and packing, shall be obtained from the same source of supply as used in regular production.

4.2.1 Government testing.- When the Government is responsible for conducting first article approval tests, the contractor, prior to submitting the sample to the Government, shall inspect the sample to insure that it conforms to all the requirements of the contract and submit a record of this inspection with the sample, including certificates of conformance for materials.

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4.2.2 Contractor testing.- When the contractor is responsible for conducting first article approval tests, the sample shall be inspected by the contractor for all the requirements of the contract. The sample and a record of this inspection, including certificates of conformance for materials, shall be submitted to the Government for approval. The Government reserves the right to witness the contractor's inspection.

4.3 Inspection provisions.

4.3.1 Submission of product.- Unless otherwise specified by the contracting officer, inspection lot size, lot formation and presentation of lots shall be in accordance with "Submission of Product" provisions of MIL-STD-105.

4.3.2 Examination and tests.

4.3.2.1 Components and subassemblies.- All components and subassemblies shall be inspected in accordance with the inspection provisions contained in Supplementary Quality Assurance Provisions (SQAP) listed in the technical data package (TDP). In the absence of SQAP's, the applicable Quality Assurance Provisions of MIL-F-13926 shall apply.

4.3.2.2 Final acceptance inspection.- Subsequent to first article approval, examination and tests related to Section 3 herein shall be performed on a single defect (individual characteristic) basis in accordance with MIL-STD-105 and the sampling plans specified in Tables III, IV, and V herein. Examination and test for packaging and marking shall be in accordance with MIL-P-14232 and Section 5 herein. The tabulated classification of defects in Tables III, IV, and V shall constitute the minimum inspection to be performed by the supplier after first article approval and prior to Government acceptance or rejection by item or lot.

4.3.2.3 Functional tests.- The requirements and tests in Table III shall be inspected on a 100% basis. All examinations and tests shall be conducted at the standard ambient temperature (60° to 90°F).

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TABLE III

(100 percent inspection)

CHARACTERISTICS	REQUIREMENT	TEST PROCEDURE
101 Malfunction 6	3.3.1.1	4.6.2
102 PFN Voltage	3.3.1.2	4.6.3
103 Fire Signal	3.3.1.3	4.6.4
104 Range Light	3.3.1.4	4.6.5
105 Current limiting of Power Driver	3.3.1.5	4.6.6

4.3.3 Acceptance and rejection.- Rejected lots shall be screened for all defective characteristics. Removal or correction of defective units and resubmittance of rejected lots shall be in accordance with "Accptance and Rejection" as specified in MIL-STD-105.

4.4 Special sampling.

4.4.1 General.- One power supply shall be selected at random by a Government representative as a control sample from each 100 produced. The control sample shall meet the requirements and tests in Table IV.

TABLE IV

CONTROL TEST	REQUIREMENT	TEST PROCEDURE
301. Fabrication	3.1	Applicable drawings - Visual
302. General Specification	3.2	MIL-F-13926 - Visual

4.4.2 Environmental.- Three power supplies shall be selected at random by a Government representative as a control sample from each 50 produced or from each month's production, whichever occurs first. The control samples shall meet the requirements and tests in Table V.

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TABLE V

CONTROL TEST	REQUIREMENT	TEST PROCEDURE
303. Storage temperature	3.3.2.1	4.6.9.1
304. Operating temperature	3.3.2.2	4.6.9.1
305 Vibration	3.3.2.3	4.6.9.2

4.4.3 Failure of sample.- Should any one item of a special sampling fail to meet the specified test requirements, acceptance of the product shall be suspended by the Government until necessary corrections have been made by the contractor and the resubmitted item has been approved (see 4.3.3).

4.5 Inspection equipment.- Except as otherwise provided for by the contract, the contractor shall supply and maintain inspection equipment in accordance with the applicable requirements of MIL-I-45607.

4.5.1 Government furnished inspection equipment.- Where the contract provides for Government furnished test equipment, supply and maintenance of test equipment shall be in accordance with the applicable requirements of MIL-I-45607.

4.5.2 Contractor furnished inspection equipment.

4.5.2.1 Government design.- Unless otherwise specified in the contract, all inspection equipment specified by drawing number in specifications or SQAP forming a part of the contract shall be supplied by the contractor in accordance with applicable technical data listed in the technical data package (TDP).

4.5.2.2 Contractor design.- The contractor shall design and supply inspection equipment compatible with the "Test Methods and Procedures" specified in 4.6 of this specification and with the component inspection procedures specified in "Examination" and "Test Facilities" requirements of MIL-F-13926. Since tolerance of test equipment is normally considered to be within 10% of the product tolerance for which it is intended, this inherent error in the test dquipment design must be considered as part of the prescribed product tolerance limit. Thus, concept, construction, materials, dimensions, and tolerances used in the design of test equipment shall be so selected and controlled as to insure that the test equipment will reliably indicate

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acceptability of a product which does not exceed 90% of the prescribed tolerance limit and permit positive rejection when nonconforming. Construction shall be such as to facilitate routine calibration of test equipment.

4.5.3 Test equipment. - In conjunction with 4.5.2, the standard test equipment listed in Table VI shall be utilized in performing the applicable test of 4.6.

TABLE VI

	NOMENCLATURE	DESCRIPTION
1.	D-C Power Supply	Plus 5 ±0.2V Ripple: 25 mV p-p max
2.	D-C Power Supply	Plus 15 ±0.7V Ripple: 25 mV p-p max
3.	D-C Power Supply	Minus 6 ±0.2V Ripple: 25 mV p-p max
4.	D-C Ammeter	Range: 0 to 15 amps Accuracy of ±2%
5.	D-C Microammeter	Range: 0 to 100 μa Accuracy of ±2%
6.	D-C Voltmeter	Range: 0 to 500 volts Accuracy of ±1%
7.	D-C Voltmeter	Range: 0 to 2000 volts Accuracy of ±1%
8.	Oscilloscope	Frequency response: dc to 500 kHz Input impedance: 1 megohm or greater Sensitivity: 5mV/cm Sweep time: 0.2 μs/cm
9.	Signal source	Plus 8 ±0.5 Vdc Source impedance: 33 kilohms ±5%

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TABLE VI (CONTINUED)

NOMENCLATURE	DESCRIPTION
10. Signal source	Plus 17 to 26 Vdc adjustable Source impedance: less than 0.01 ohm
11. Signal source	Capable of providing dc level of amplitude specified in item 3.1, Table I.

4.6 Test methods and procedures.

4.6.1 Test conditions. - Unless otherwise specified, the test conditions shall be in accordance with the "Test Facilities" requirements of MIL-F-13926.

4.6.1.1 Initial conditions. - Connect the loads, power and signals as specified in Table I.

4.6.2 Malfunction 6. -

(a) Apply the logical zero type-A signal of Table I to P1-4 (P1-9 return). The voltage output at P1-8 shall meet the requirements of 3.3.1.1.
(b) Connect the 66.5 kilohm load of Table I between P1-5 and P1-15. The voltage output at P1-8 shall meet the requirements of 3.3.1.1.

4.6.3 PFN voltage. - Apply the logical zero type-A signal of Table I to P1-2, and a momentary logical zero type-A signal to P1-3 and P1-4, and perform the following tests.

4.6.3.1 PFN voltage output. - Connect the 24 Vdc of Table I, adjusted to 24 ±1 Vdc, to P1-19 and P1-20, with the negative returns to P1-17 and P1-18. The PFN output voltage at P1-A-1 shall meet the requirements of 3.3.1.2.1.

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4.6.3.2 Dump relay. - Adjust the PFN voltage at P1-A-1 to 1200 ±100 Vdc using R10. Remove the 24 Vdc from P1-19 and P1-20. The PFN output voltage at P1-A-1 shall meet the requirement of 3.3.1.2.2.

4.6.3.3 PFN charge time. -

(a) With R10 set as in 4.6.3.2, reapply the 24 Vdc of Table I, adjusted to 24 ±1 Vdc, to P1-19 and P1-20. The voltage and charge time of P1-A-1 shall meet the requirements of 3.3.1.2.3.
(b) Remove the 24 Vdc from P1-19 and P1-20, adjust the level to 18.5 ±0.5 Vdc, and reapply the 18.5 Vdc to P1-19 and P1-20. The voltage and charge time at P1-A-1 shall meet the requirements of 3.3.1.2.3.

4.6.3.4 Repeaking inhibit. -

(a) Connect the 8 Vdc of Table I to P1-7. The PFN voltage at P1-A-1 shall meet the requirements of 3.3.1.2.4.
(b) Remove the 8 Vdc from P1-7 and change the logical zero type-A signal at P1-2 to logical one. The PFN voltage at P1-A-1 shall meet the requirements of 3.3.1.2.4.
(c) Change the logical one type-A signal at P1-2 to logical zero, and connect the 66.5 kilohm load of Table I to P1-5 and P1-15. The PFN voltage at P1-A-1 shall meet the requirements of 3.3.1.2.4.

4.6.3.5 PFN voltage range adjustment. - Vary the PFN voltage output at P1-A-1 by adjusting R10. The PFN voltage at P1-A-1 shall meet the requirements of 3.3.1.2.5.

4.6.3.6 PFN voltage programming. -

(a) Adjust the PFN voltage at P1-A-1 to 1200 ±15 Vdc using R10, remove the 100 kilohm load of Table I from P1-6 and P1-9, and connect the 32.4 kilohm load to P1-6 and P1-9. The PFN voltage at P1-A-1 shall meet the require requirements of 3.3.1.2.6.

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(b) Replace the 32.4 kilohm load with the 3.32 megohm load of Table I. The PFN voltage at P1-A-1 shall meet the requirements of 3.3.1.2.6.

4.6.3.7 15-Vdc interlock. - With the voltage at P1-A-1 initially zero and the 15 Vdc removed from P1-15, apply the 24 Vdc to P1-19 and P1-20. The PFN charge circuit condition shall be as specified in 3.3.1.2.7.

4.6.3.8 400-volt output. - With the 15 Vdc of Table I applied as specified, apply the 24 ±1 Vdc to P1-19 and P1-20. The output voltage at P1-10 shall meet the requirements of 3.3.1.2.8.

4.6.3.9 PFN voltage sensing. - Adjust the PFN voltage at P1-A-1 to 1200 ±25 Vdc. The current through a microammeter connected between P1-5 and P1-9 shall meet the requirements of 3.3.1.2.9.

4.6.4 Fire signal. -

4.6.4.1 Apply the logical zero type-A signal of Table I to P1-2, a momentary logical zero, type-A signal to P1-3 and P1-4, and the 24 ±1 Vdc of Table I to P1-19 and P1-20. Adjust the PFN voltage at P1-A-1 to 1200 ±200 Vdc. The output voltage at P1-11 shall meet the requirements of 3.3.1.3.1.

4.6.4.2 With the PFN output voltage and P1-A-1 still adjusted to 1200 ±200 Vdc, apply the inputs of Table II to the pins specified. The output voltages at P1-11 shall meet the requirements of 3.3.1.3.2.

4.6.5 Range light. -

(a) Apply the momentary logical zero, type-A signal of Table I to P1-3. The voltage output at P1-14 shall meet the requirements of 3.3.1.4.1.

(b) Apply the logical zero, type-A signal of Table I to P1-2, a momentary logical zero, type-A signal to P1-4, and the 24 ±1 Vdc to P1-19 and P1-20. The waveform at P1-14 shall meet the requirements of 3.3.1.4.2.

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(c) Apply the momentary logical zero type-A signal of Table I to P1-1. The voltage output at P1-14 shall meet the requirements of 3.3.1.4.3.

4.6.6 Current limiting of power driver. -

(a) Apply the momentary logical zero type-A signal of Table I to P1-3 and P1-4.
(b) Connect a short-circuit between P1-A-1 and P1-9.

CAUTION: Discharge the capacitor load between P1-A-1 and P1-9 before applying the short-circuit. The short-circuit condition shall be limited to 5 seconds.

The average current of the 24 Vdc signal applied to P1-19 and P1-20 shall meet the requirements of 3.3.1.5.

4.6.7 Environmental tests. - The test equipment shall be in accordance with the "Test Facilities" requirements of MIL-F-13926 except that the power supply may be mounted on an inside surface of the test chamber for the temperature test and the tolerance requirements for vibration frequency measuring instruments shall not apply.

4.6.7.1 Temperature test. -

NOTE: Before performing the temperature cycling, adjust the PFN output voltage at P1-A-1 to 1200 ±15 Vdc. The rate of temperature change shall be less than 7°F per minute during temperature cycling.

(a) Place the power supply in the test chamber and reduce the chamber temperature to -80°F.
(b) Soak the power supply at this temperature for at least 5 hours with the unit not operating.
(c) Raise the chamber temperature to -40°F.
(d) Following thermal stabilization at -40°F, place the power supply in operation and perform the tests specified in 4.6.2, 4.6.3, 4.6.4, 4.6.5, and 4.6.6.

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(e) Raise the chamber temperature to +160°F.
(f) Soak the power supply at +160°F for at least 5 hours with the power supply not operating.
(g) Reduce the chamber temperature to +125°F.
(h) Following thermal stabilization at +125°F, place the power supply in operation and perform the tests specified in 4.6.2, 4.6.3, 4.6.4, 4.6.5, and 4.6.6.
(i) Reduce the chamber temperature to ambient.
(j) Following stabilization at ambient temperature, place the power supply in operation and perform the test specified in 4.6.2, 4.6.3, 4.6.4, 4.6.5, and 4.6.6.

The power supply shall meet the requirements of 3.3.2.1 and 3.3.2.2.

4.6.4.2 Vibration. - With the power supply firmly mounted to the vibration table perform the following:

(a) Test in accordance with Method 514-I of MIL-STD-810 for category (f) equipment, except that the acceleration shall not exceed ±4g and the maximum excursion (double amplitude) shall be 0.5 inch.
(b) Sweep the frequency from 5 to 500 Hz and back to 5 Hz in a period of 15 minutes. Perform two sweeps in each of the three mutually perpendicular axes.
(c) Following vibration cycling, examine the power supply for structural failures, loose hardware, wire breakage and abrasion.
(d) Place the power supply in operation and perform the tests specified in 4.6.2, 4.6.3, 4.6.4, 4.6.5 and 4.6.6.

The power supply shall meet the requirements of 3.3.2.3.

5. PREPARATION FOR DELIVERY

5.1 Packaging, packing and marking. - Packaging, packing and marking shall be in accordance with Packaging Data Sheet 10559345. The level of protection shall be as specified in the procurement document.

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6. NOTES

6.1 Intended use.- The PFN Charge Power Supply is a sub-assembly of the Laser Receiver-Transmitter, intended for use in the Laser Range Finder AN/VVS-1. Its primary function is to charge the PFN (pulse forming network) to the required voltage from a 28-volt supply and to maintain this charge until the system is fired. Secondary functions include over-voltage sensing, over-current protection, and generation of the Q-switch motor turn-on signal and malfunction signal.

6.2 Ordering data.- Procurement documents should specify the following:

(a) Title, number, and date of this specification.
(b) Applicable packaging data sheet number (see 5.1).
(c) Selection of applicable levels of preservation, packaging and packing
(d) Applicable stock number.
(e) Requirement for First Article Submission.

Custodian:

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Preparing activity:

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